University of Westminster
School of Electronics and Computer Science

6MMCS009W Usability Testing and Evaluation Specification (2025/26)

Module leader	Dr Daphne Economou
Unit	Individual Coursework [CW2]
Weighting	Individual Coursework [CW2] 50%
Qualifying mark	30%
Description	**Individual coursework** [CW2] - assesses the students' skills in: using statistical packages to organise data that has been collected to address a usability testing research question; selecting appropriate statistical methods to analyse the data; reporting usability testing findings and making recommendations to improve the usability of a digital product.
Learning Outcomes Covered in this Assignment	**Individual Coursework [CW2]** LO4 Apply suitable analytical and/or statistical methods to analysis and interpret results to assess the design; LO5 Formulate and make recommendations to inform decision-making in terms of design and development of the product.
Handed Out	20th October 2025
Due Date	**Individual Coursework [CW2] – 7th January 2026, 1pm Online**
Expected deliverables	**Individual Coursework [CW2]** • A report containing all the results, the statistical analysis, reports, graphs related to the hypothesis you have formed and you evaluate • SPSS or PSPP files with organised data
Method of Submission	Electronic submission on Blackboard
Type of Feedback and Due Date	**Formative Feedback:** • Present progress on the Individual Coursework [CW2] to your tutor/s by weeks 11 & 12 during tutorial/seminar time to get instant formative feedback and update your report before the final submission **Summative feedback (marks and written feedback)** on both CW will be provided within 3 weeks after the submission (mark on Blackboard, additional oral feedback in class and discussion of model answers). All marks will remain provisional until formally agreed by an Assessment Board.

Coursework assessment regulations

Refer to Section 6 Coursework Assessment Regulations for taught courses guide for undergraduate students for a clarification of how you are assessed, penalties and late submissions, what constitutes plagiarism etc.

https://www.westminster.ac.uk/sites/default/public-files/general-documents/Academic-regs-2025-26-Section-6-%20Coursework-regulations.pdf

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark, as a penalty for late submission, except for work which obtains a mark in

the range 40 – 49%, in which case the mark will be capped at the pass mark (40%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognised that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Campus Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regarding University Assessment Regulations, please refer to the following website: https://www.westminster.ac.uk/sites/default/public-files/general-documents/Academic-regs-2025-26-Section-6-%20Coursework-regulations.pdf

Individual Coursework description [CW2]

An **e-learning platform** is redesigning its course player to improve **task efficiency** and **learner satisfaction**. A UX team produced **four layout variants** (A–D), each available in **three colour themes** (Light/Mid/Dark) and **three base font sizes** (Small/Medium/Large). Prototypes were tested on **three device types** (Phone/Tablet/Desktop) with real users completing core tasks (e.g., find lesson notes, change playback speed, mark lesson complete).

You are provided with a table that contains the study results (see the raw data spreadsheet) for all relevant UX metrics. Select the appropriate columns that align with a research question and hypothesis you will formulate. Then, perform a statistical analysis of the data to evaluate these hypotheses and compile your findings in a well-organized report.

Describe the process you followed in a report that details your statistical approach, results, and evidence-based conclusions to guide design recommendations for the client.

Form **two hypotheses** that can be evaluated using the provided raw data and complete the following tasks:

1. **Hypothesis Formation, Statistical Method Selection, and Rationale**
    - Define two research questions and state the corresponding hypotheses you aim to test. Identify the specific columns in the dataset you have been given that will be used to analyse each hypothesis.
    - Justify the selection of appropriate statistical methods by choosing two from the following options and explain why each method aligns with the data structure and your research objectives:
        - Independent-sample t-test or z-test
        - Paired-sample t-test or z-test
        - ANOVA
        - Pearson correlation
    - Provide the mathematical expressions for each hypothesis, define the variables involved, and outline the experimental design (e.g., how participants or groups are organized relative to the hypotheses).
2. **Organize Data in a Statistical Package**
    - Use a statistical package (e.g., SPSS or PSP) to organize the relevant columns from the raw data for efficient analysis.
    - In SPSS, set up appropriate variables in the "Variable View," ensuring each is correctly labelled and defined according to analysis requirements.
    - Arrange the data in the "Data View" in a way that aligns with your chosen statistical methods, including creating any necessary computed variables or transformations to meet analytical requirements. Clean the data if needed by addressing missing, outlying, or inconsistent entries to ensure data integrity.
3. **Conduct Descriptive Statistics**
    - Calculate and record descriptive statistics, including measures of central tendency (mean, median, and mode) and variability (range, variance, and standard deviation).
    - Organize and present these results in a clear, labelled table, summarizing these statistics for each variable to give an overview of key data characteristics.
4. **Evaluate Data Distribution**
    - Choose and produce appropriate graphical representations for the data (e.g., histograms, box plots, pie charts, or bar charts) and justify each choice. Provide clear descriptions for each graph.

- Assess the normality of the data by reviewing descriptive statistics and analyzing the graphical representations. Based on these normality assessments, confirm whether the selected statistical methods are appropriate for each hypothesis.

5. **Hypothesis Testing with Statistical Analysis**
    - Conduct two of the following statistical tests that correspond to your hypotheses:
        - Independent-sample t-test
        - Paired-sample t-test
        - ANOVA
        - Pearson correlation to assess relationships between variables
    - Present the results.
6. **Report and Interpret Results**
    - Summarize and interpret all results based on descriptive statistics, confidence intervals, and p-values for each hypothesis.
    - Use these tests to evaluate each hypothesis and record the results.
7. **Usability Analysis and Recommendations**
    - Based on your analysis, write a brief discussion on the overall usability of the application. Identify any usability issues, and provide recommendations on design areas that could be improved to enhance the user experience. Ensure the report is structured, formal, and uses clear language.

Report Writing and Data Presentation

Present all your analysis in a form of a report detailing the entire process of creating and conducting the usability test. Include a critical reflection on the overall process, discussing successes, challenges, and areas for improvement.

Follow the provided template.

The report should be approximately 3000 words excluding the appendix, although this is not a strict word count.

Individual Coursework submission [CW2]

Submit individually:

- A report containing all the results, the statistical analysis, reports, graphs related to the hypotheses you have formed and you evaluate
- SPSS or PSPP files with organised data

The expected report structure is as follows:

- A cover page with your student name and ID
- Explain clearly and in detail all the above stages related to the statistical analysis of the data and the evaluation of the hypothesis. Produce and discuss the right graphical results (e.g. pie-chart, bar-chart, histograms, box-plots), provide a justification about the chosen graphical representation, and identify any large departures (correct selection and appropriate discussion of graphs). Base on the analysis of the hypotheses you have formed write a short discussion about the overall usability of the application. Identify usability issues and recommend which areas on the design should be changed to improve the user experience. Pay attention on the language, structure and formality of the report.

Formative feedback

Take the opportunity to present the Individual Coursework [CW2] to your tutor/s by weeks 11 & 12 during tutorial/seminar time to get instant formative feedback and update your report before the final submission.

Final Marking and feedback

Provisional marks will be provided within 15 working days from the submission deadline. Marks are finalised at the exam board.

Individual Coursework [CW2] marking scheme

This assignment is marked based on the following marking criteria.

	Individual coursework [CW2] requirements	mark
1a	**Hypothesis 1 Formation, Statistical Method Selection, and Rationale**. Correct research question (1 mark), correct hypothesis (1 mark), correct mathematical expression of the hypothesis (1 mark), correct definition of the variables involved (1 mark), and correctly outline the experimental design (1 mark).	5
2a	**Organise the collected data for hypothesis 1 using a statistical package (SPSS or PSP),** set up appropriate variables in the "Variable View," (2 marks), ensuring each is correctly labelled and defined according to analysis requirements (2 marks). Arrange the data in the "Data View" in a way that aligns with your chosen statistical methods (3 marks), including creating any necessary computed variables or transformations to meet analytical requirements (1 mark). Clean the data if needed by addressing missing, outlying, or inconsistent entries to ensure data integrity (1 mark).	9
3a	**Conduct Descriptive Statistics for hypothesis 1 .** Calculate and record descriptive statistics, including measures of central tendency (mean, median, and mode) and variability (range, variance, and standard deviation) (4 marks). Organize and present these results in a clear, labelled table, summarizing these statistics for each variable to give an overview of key data characteristics. (2 marks)	5
4a	**Evaluate Data Distribution for the dataset used for hypothesis 1 .** Choose and produce appropriate graphical representations for the data (e.g., histograms, box plots, pie charts, or bar charts) and justify each choice (2 marks). Provide clear descriptions for each graph (4 marks). Assess the normality of the data by reviewing descriptive statistics and analyzing the graphical representations confirm whether the selected statistical methods are appropriate for each hypothesis (3 marks).	9
5a	**Hypothesis 1 Testing with Statistical Analysis.** Conduct two of the following statistical tests that correspond to your hypotheses: - Independent-sample t-test - Paired-sample t-test - ANOVA - Pearson correlation to assess relationships between variables Present the results	**5**
6a	**Report and Interpret Results for hypothesis 1.** Summarize and interpret all results based on descriptive statistics (2 marks), confidence intervals (2 marks), and p-values for each hypothesis (2 marks). Use these tests to evaluate the hypothesis and record the results (2 marks).	**8**
1-6b	**Same as above for hypothesis 2**	41
7	Base on the analysis of the hypotheses you have formed write a short **discussion about the overall usability** (4 marks) of the application (critical discussion of the results based on the analysis) (4 marks)	8
8	**Report Writing and Data Presentation** (Clarity and Organization, References and Citations, Adherence to Guidelines (2 marks), Completeness and Detail (2 marks), Use of Evidence and Data Presentation (2 marks), Critical Analysis and Reflection (2 marks), Writing Quality and Style (2 marks)). **Not removing instructions from the report (- 4 marks)**	10
	CW2 Total	**100**